

ARAVIND
SIDDHARTH

ROBOTICS ENGINEER

siddhartharavind8@gmail.com

+91 9360764454

No:2 Raman Thottam 2nd Street, Nanganallur,
Chennai-600114.

linkedin.com/in/aravind-siddharth-52a496231

github.com/Aravindsiddharh

PROFILE

Dedicated and motivated Bachelor of Technology candidate in Electrical and Electronics Engineering, with a strong foundation in Robotics and Control Systems. Skilled in designing and tuning feedback control algorithms (PID, state-space), dynamic modeling, and trajectory tracking for mobile robots. Proficient in Robot Operating System (ROS), simulation platforms like Gazebo and CoppeliaSim, and tools such as MATLAB and Simulink. Eager to contribute to innovative advancements in autonomous systems, AI integration, and robotic control applications.

SKILLS

ROS	
Gazebo	
MATLAB	
ARDUINO	
Raspberry Pi	
Python	
C++	
OpenCV	
Fusion360	
Ansys Workbench	
Jetson orin	

EDUCATION

Bachelor of Technology,
SASTRA UNIVERSITY
10/2020 – 11/2024 | Thanjavur, India

XII Grade
The Hindu Colony Chellammal Vidyalaya
04/2018 – 03/2022 | Chennai, India

PROJECTS

A Butler Robot for Restaurant operation
11/2024 – 12/2024

This project is designed for restaurant operations, specifically automating tasks like food delivery from the kitchen to customers' tables. It manages multiple orders efficiently, ensures accurate navigation in a dynamic environment, and handles scenarios like cancellations and timeouts. By integrating ROS for task management, path planning, and obstacle avoidance—alongside control systems for trajectory tracking and feedback-based motion regulation—the robot maintains stability and responsiveness in real time. These enhancements streamline restaurant operations, reduce human effort, and improve service efficiency

Autonomous navigation for tugged vehicles in ROS2 using nav2.
07/2023 – 11/2023

This project aims to develop an advanced autonomous navigation system for an Autonomous Mobile Robot (AMR) tailored specifically for warehouse operations. The system utilizes state-of-the-art Simultaneous Localization and Mapping (SLAM) techniques alongside the nav2 tools within the Robot Operating System (ROS). Integrated control systems such as PID-based velocity and heading controllers ensure smooth and accurate motion, especially in dynamic environments. This cohesive setup enables the AMR to navigate autonomously, dynamically map its surroundings, and maintain precise real-time localization. The project's ultimate goal is to optimize warehouse logistics by improving efficiency, reliability, and safety through robust navigation and motion control capabilities.

Autonomous navigation of Turtlebot3
03/2021 – 04/2021

This project focuses on implementing autonomous navigation using the TurtleBot3 platform. Utilizing ROS (Robot Operating System) and its navigation stack, the goal is to enable the TurtleBot3 robot to navigate autonomously in indoor environments. Key objectives include developing and tuning navigation algorithms, integrating sensors for obstacle avoidance, and optimizing path planning for efficient traversal. The project aims to demonstrate the feasibility and effectiveness of autonomous robotics in real-world applications, contributing to advancements in robotic navigation.

PROFESSIONAL EXPERIENCE

Doozy Robotics India Pvt ltd
Robotics Engineer
06/2025 – Present | Chennai, India

- Implemented **Route Server** for graph-based path planning and route execution using **ROS 2 Nav2**.
- Developed **dynamic obstacle avoidance** during route operations by integrating with custom global planner and behavior tree conditions.

PUBLICATIONS

AI-Based Vision System for Autonomous Navigation in a Self-Driving Toy Car

2024

- Developed a self-driving toy car utilizing AI and computer vision. A webcam captured track data, processed into driving commands (right, left, forward, stop) using a Convolutional Neural Network (CNN). The CNN predictions were transmitted to an Arduino for motor control via serial communication, enabling precise navigation without additional sensors. This study highlights cost-effective, scalable approaches to autonomous systems, advancing intelligent transportation technologies.
- Keywords: Artificial intelligence, computer vision, autonomous navigation, CNN, self-driving systems

EXTRACURRICULAR ACTIVITIES

Proficient in Prompt Engineering for AI-related tools, expanding knowledge in this specialized field.

Engaged in volleyball, honing teamwork and coordination skills essential for collaborative environments.

Achieved third place in state-level chess tournament.

COURSES

Course completed on Modern Robotics, Course 1: Foundations of Robot Motion in coursera

04/2025

Courses completed on ROS Basics Motion Navigation by Anis Koubaa Udemy.

12/2024

Course completed on Robotics and automation conducted by the Ministry of Heavy Industries.

04/2023 – 12/2023

Participated in workshops conducted by RCS(Robotics Club Sastra) COBOTS

12/2022 – 01/2023

LANGUAGES

Tamil ● ● ● ● ●

English ● ● ● ● ●

German ● ● ● ● ●

- Created **custom global planner** and **local controller plugins** for Nav2 to optimize navigation paths and motion control.
- Designed and implemented **pause/resume functionality** for ongoing navigation tasks via ROS actions and lifecycle management.
- Built a package for re-localization of a robot using April-Tag when it is lost in map
- Built a **multi-robot navigation and fleet management system**, enabling coordinated task allocation and route execution for multiple kind of robots.

Indian Institute of Technology, Madras

Design of Modular and Reconfigurable AMRs for Human Assistance

01/2024 – 07/2024

- Focused on designing modular AMRs for human assistance, aiming to enhance adaptability and efficiency in dynamic environments.
- Conceptualised architectures, prototyping configurations, and evaluating performance, enriching expertise in robotics and human-centric design for future assistive technology advancements.

YASKAWA India Pvt Ltd

Robotics Intern

07/2023 | Bengaluru, India

- Conducted a comprehensive study on robotic manipulators, exploring various configurations and design architectures suited for different tasks.
- Analyzed forward and inverse kinematics to understand motion planning and workspace optimization. Investigated control mechanisms including PID, adaptive, and model-based control strategies for precise motion.
- Examined practical applications and limitations, emphasizing efficiency, payload capacity, and end-effector versatility.

BSNL

Network Intern

07/2022 – 08/2022 | Chennai, India

- Supported network operations by configuring and troubleshooting routers, switches, and firewalls, while also assisting in network monitoring and documentation tasks.
- Gained hands-on experience in TCP/IP networking, protocol analysis, and network security fundamentals under the mentorship of senior network engineers.

IBM India Pvt Ltd

Robotics Intern

04/2021 – 05/2021 | Online

- Collaborated on developing autonomous navigation algorithms for mobile robots, contributing to the integration of computer vision and sensor fusion techniques.
- Applied theoretical knowledge in a practical setting, gaining insights into real-world robotics applications and enhancing skills in algorithm development and implementation